

AI FOR GAME ARTISTS & MODDERS

# PixelForge

Old game textures in.  
**4× HD out.**

4× upscale

Trained on game art

Runs in browser

Before (bicubic)



PixelForge 4×



## THE PROBLEM

# HD screens expose every old texture.

1

### Remasters need thousands of textures

Redrawing every wall, floor and prop by hand is the slowest, most expensive part of a remaster or HD mod.

2

### Simple resizing just blurs

Stretching a texture 4× with bicubic gives soft, muddy surfaces that look worse on a modern monitor.

3

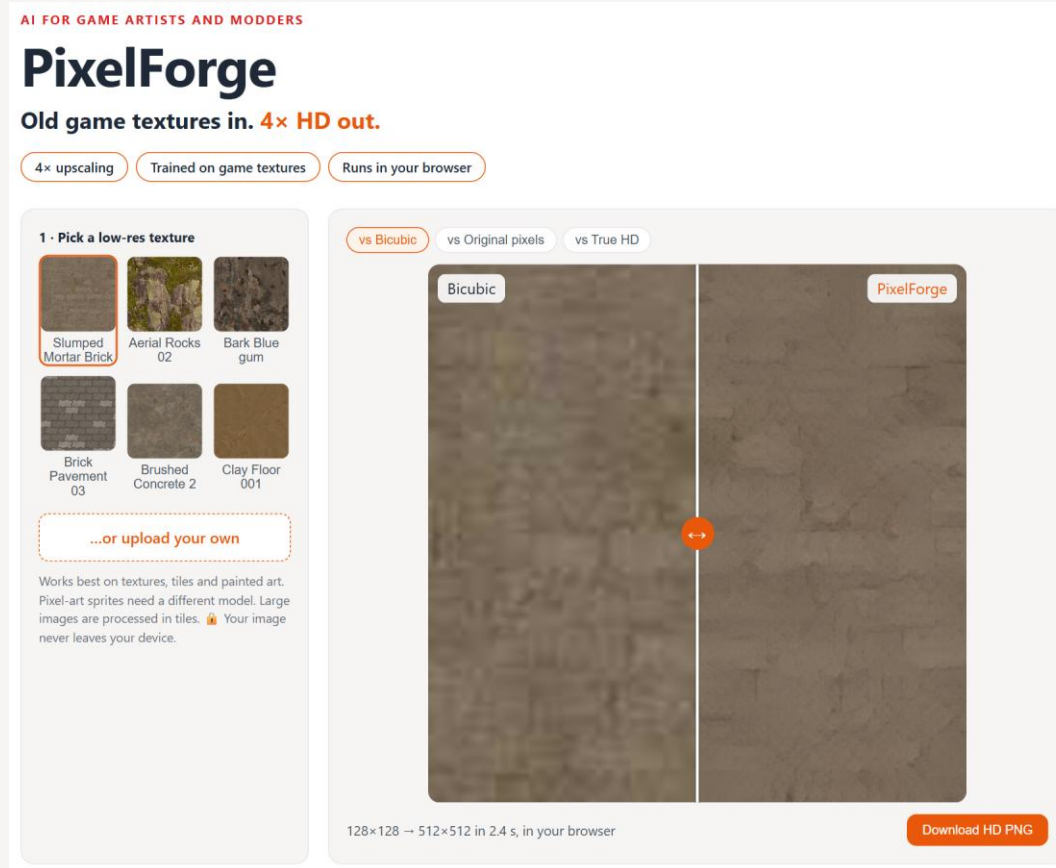
### Generic AI upscalers aren't tuned for games

Tools trained on photos can smear tiling patterns or invent the wrong kind of detail.

PixelForge does the first pass for you: **seconds per texture, in your browser.**

## TRY IT YOURSELF

# Drop in a texture. Drag the slider.



1

### Before/after slider

Compare against bicubic, the original pixels, or the true HD texture.

2

### Runs in your browser

No upload, no server, no install. Your art never leaves your machine.

3

### Download the HD file

Large images are processed in tiles; get a 4x PNG in seconds (2.3 s for a 512x512 result in a laptop browser).

## RESULTS

# Looks like real HD. Small enough for a browser.

### 41%

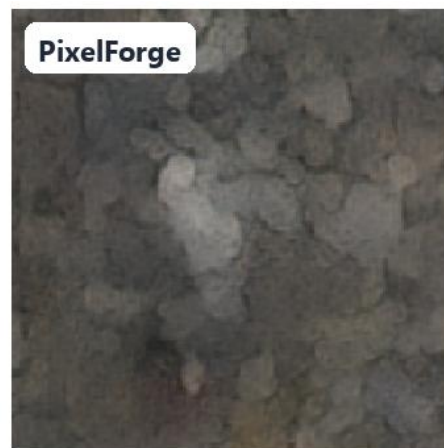
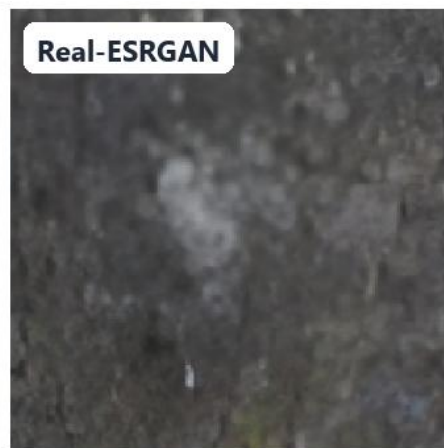
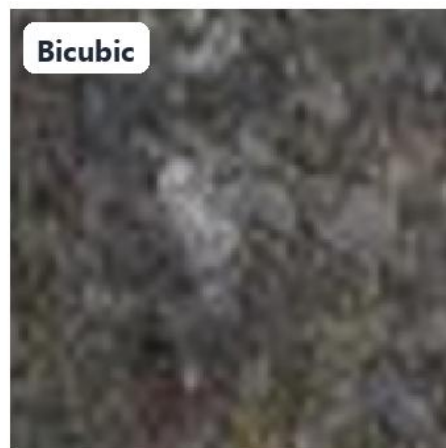
closer to the real HD look than bicubic  
LPIPS perceptual distance, 60 unseen  
game textures

### 15× smaller

than Real-ESRGAN (1.1M vs 16.7M  
parameters)  
LPIPS 0.414 vs 0.454 (lower is better)

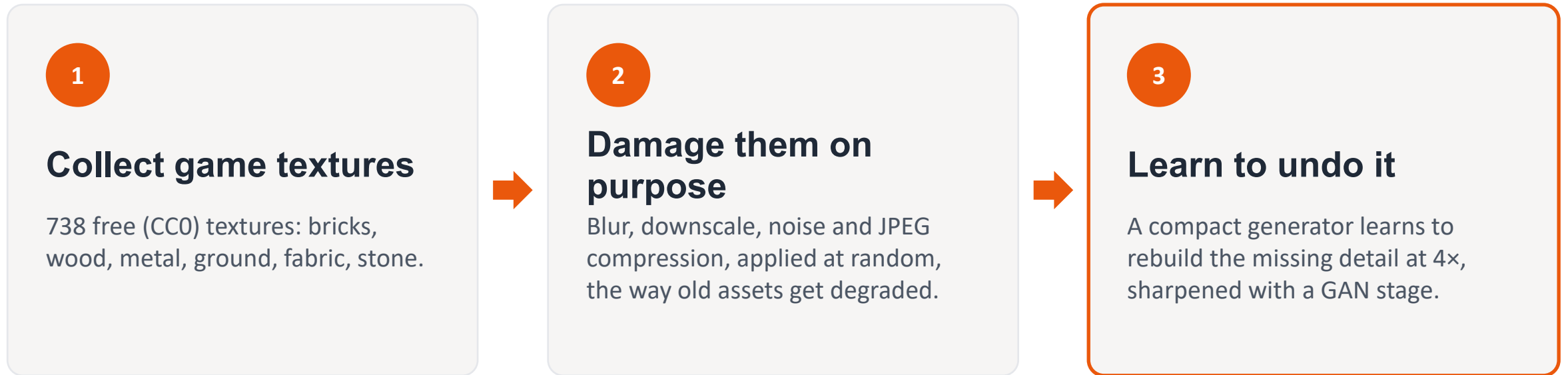
### 2.3 s

to upscale a texture 4× in a laptop  
browser  
128×128 → 512×512, no GPU, no upload



## HOW IT WORKS

# Trained to undo old-texture damage.

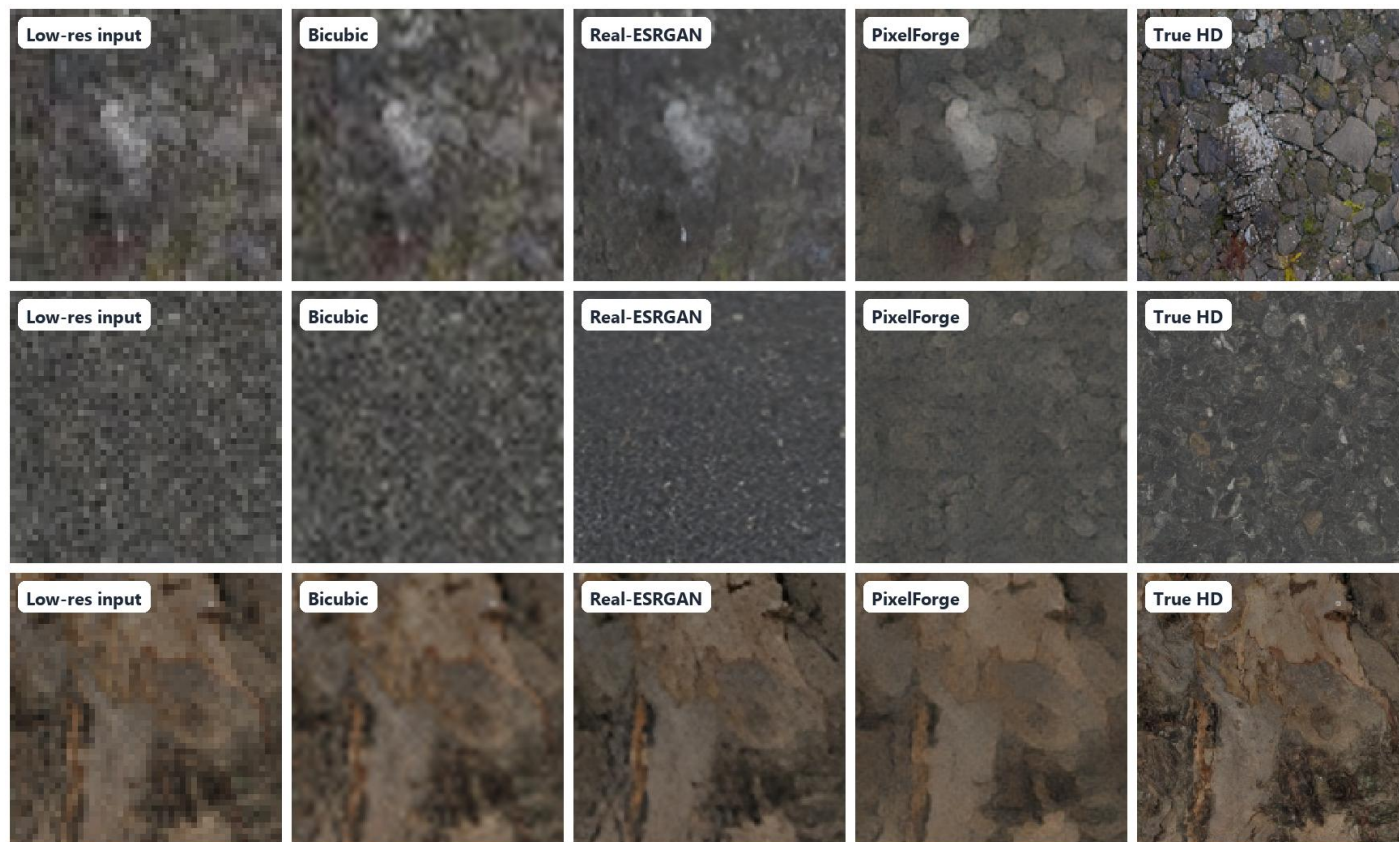


**For engineers:** compact ESRGAN-style RRDB generator (1.1M params) trained from scratch in PyTorch: L1 stage, then VGG-perceptual + U-Net-discriminator GAN stage with EMA (Real-ESRGAN recipe), exported to ONNX. Benchmarks are in the appendix.

## REAL RESULTS

# Real results on textures it never saw.

Median-improvement textures, different materials, zoomed. PixelForge clears noise and JPEG blocks; detail the tiny input no longer has can't be fully recovered.





## HOW WE'D WORK TOGETHER

# Start with a test pack. Scale to the whole library.

### STEP 1

#### Send 10–20 textures

A representative sample of your game's textures or asset pack.

### STEP 2

#### Test pack back

I upscale them, optionally fine-tune on your art style, and send before/after comparisons.

### STEP 3

#### Batch the library

Folder in, folder out, file names kept, ready to drop into your mod or engine.

**What you get:** the HD texture set · a batch tool you can rerun on new assets · optional model fine-tuned on your art style · before/after report

NEXT STEP

**Send me 10 of your textures.**  
**I'll send them back in HD.**



**Try the live demo**

[priyanshu-singh-git.github.io/pixelforge/](https://priyanshu-singh-git.github.io/pixelforge/)

**Code and full benchmark**

[github.com/Priyanshu-Singh-git/pixelforge](https://github.com/Priyanshu-Singh-git/pixelforge)



# The numbers behind it.

Held-out test set: 60 game textures, realistic degradation, ×4. PSNR/SSIM higher is better; LPIPS lower is better.

| Method                              | Params       | PSNR ↑       | SSIM ↑       | LPIPS ↓      | GPU ms      |
|-------------------------------------|--------------|--------------|--------------|--------------|-------------|
| Bicubic                             | —            | 29.31        | 0.612        | 0.707        | —           |
| Lanczos                             | —            | 29.24        | 0.608        | 0.711        | —           |
| Real-ESRGAN x4plus (pretrained)     | 16.70M       | 27.76        | 0.545        | 0.454        | 185.4       |
| Real-ESRGAN general-v3 (pretrained) | 1.21M        | 28.93        | 0.597        | 0.547        | 26.9        |
| <b>PixelForge SRVGG-S, L1 ★</b>     | <b>0.16M</b> | <b>29.66</b> | <b>0.617</b> | <b>0.636</b> | <b>15.2</b> |
| <b>PixelForge RRDB-S, L1 ★</b>      | <b>1.12M</b> | <b>29.73</b> | <b>0.620</b> | <b>0.659</b> | <b>39.0</b> |
| <b>PixelForge RRDB-S, GAN ★</b>     | <b>1.12M</b> | <b>28.76</b> | <b>0.571</b> | <b>0.414</b> | <b>36.0</b> |

- Architecture: the 7× smaller SRVGG-S (0.16M) gets within 0.07 dB PSNR of RRDB-S (1.12M) and slightly better LPIPS (0.636 vs 0.659), at 15 vs 39 ms: a strong candidate for real-time use.
- L1 vs GAN: the GAN stage cuts LPIPS from 0.659 to 0.414 (looks far more real) at a cost of 0.97 dB PSNR, the classic fidelity vs realism trade-off.
- Versus Real-ESRGAN x4plus (trained on photos): LPIPS 0.414 vs 0.454, PSNR 28.76 vs 27.76 dB, with 15× fewer parameters.
- Single training run per model; synthetic degradations; pixel art and alpha channels not yet evaluated.